VINCENT MORIER-GENOUD

BIOMEDICAL ENGINEER

31 years old, Swiss

Plaisance 12
1400 Yverdon-les-Bains

+41 79 722 86 53

vincent.m.genoud@gmail.com

/vincentmorier-genoud

Portfolio

Skills

Programming

Python · MATLAB · Git · C · HTML
SQL (SQLite, BigQuery) · GCP · XML

Design & Simulation

CAD & FEM (Siemens NX, Abaqus)
3D Printing (3DSlicer)

Tools & data

MS Office 365 (Excel VBA) · MS Project
Data Dashboards (Tableau, Thingsboard)
Image Analysis (FIJI)
Technical Illustration (Adobe Suite)

Soft Skills

Strong communicator · Organised
Autonomous · Curious · Teaching skills

Education

2025 **Master of Biomedical Engineering with a Major in Biomechanics**
GPA: 5,36/6
Thesis grade: 6/6
Institution: UNIBE, Bern
Master Thesis at: EPFL, Lausanne

2018 **Bachelor of Science in Automotive Engineering & *Orientation Management* Certificate**
BFH, Biel

2013 **Vocational Education and Training in Automotive Mechatronics**
ETML, Lausanne

Langues

French	Native
English	Fluent (C1)
German	Intermediate (B2)

Profile

Biomedical engineer with a background in automotive engineering, I combine over five years of experience in project management and technical development.
I have contributed to projects ranging from assistive mobility to signal processing and medical data analysis. I am now looking to bring this dual expertise to innovative projects in medical technology.

Work Experience

Master Thesis – Medical Data Scientist — 2025
Laboratory of Hemodynamics and Cardiovascular Technology, EPFL

Key Contributions: creation of a dataset from intensive care unit patients; design of signal processing and machine learning methods on the collected data; creation of comprehensive technical documentation.

Developed Expertise: measurements and instrumentation in intensive care; relational databases (SQL/BigQuery); data and signal processing in Python (NumPy, Pandas, SciPy); machine learning model development (scikit-learn, TensorFlow/Keras); certified in anonymized data management (CITI Program).

Assistive Mobility Development Engineer — 2021-2024
SCI-Mobility Laboratory, BFH Biel

Key Contributions: execution and management of engineering projects in assistive mobility; supervision of technical teams; regular progress reporting to management; acquisition and commissioning of laboratory instrumentation, including user training; implementation of a user integration approach involving people with disabilities throughout projects; management of laboratory communication.

Developed Expertise: project management (Gantt, Scrum, decision matrix); risk analysis (APR, FMEA); technical documentation in compliance with industry standards; design of test benches and static/dynamic measurements on vehicles (strain gauges, sensors); creation of comparative studies and safety tests on assistive mobility systems; participation in R&D projects; experimental data analysis on tetraplegic patients in rehabilitation.

Research Assistant — 2018-2021
Institute for Energy and Mobility Research, BFH Biel

Key Contributions: collection and analysis of data from electric vehicles (data buses, sensors); technical documentation writing on automotive communication protocols; prototype development (smartphone application, automotive and agronomic measurement tools); teaching support.

Developed Expertise: microcontroller programming (Python) and smartphone application development (Java, SQLite, XML); data transmission via telemetry (MQTT, LoRaWAN, BLE); automotive signal data analysis and processing; development of data visualization dashboards.

Volunteering

Since I was 18, I have been actively engaged in associations, developing strong organizational skills in transdisciplinary settings:

Repris de Justesse Foundation — 2013-2022
Concert musician and technical support in charge of music stage set-up

Treasurer of the Swiss Dodgeball Federation — 2018-2020
In charge of managing the accounts, I also actively organized events, such as the 2020 European centre championship in Lausanne.